

Instructions for use of ABC++

ABC++ consists of two programs, one simulates Chua's Oscillator with a piecewise linear nonlinearity, and the other simulates the circuit with a cubic nonlinearity.

To run the piecewise linear version, make sure you are in the ABC directory and then type:

```
abc(0)
```

To run the cubic version, type

```
abc(1)
```

Then, press 'Load File' to load a set of circuit parameters. The piecewise linear version comes with a vast array of such files, while the cubic version presently only has two - a-cubic-1Mhz.abc and a-cubic-1.abc. If you change the parameters, hit 'Recalculate', and the program will re-simulate the circuit for the new parameters. To save your new parameters, press 'Save File'.

Poincaré Analysis Tool

The Poincaré section tool works with the piecewise linear version. After loading a file, or simulating a set of circuit parameters, Press the 'Poincare' button in the bottom right hand corner. This will load the poincaré section tool.

The plane used to cut the trajectory is given in the form $ax + by + cz + d$, where x,y and z stand for the component of the plane in the I_3, V_2 , and V_1 axes respectively. By varying a, b, c and d , and pressing the 'Recalculate' button, the

plane will move. The 2-D map is in the bottom right hand corner. The plane may be viewed from different angles by pressing the radio buttons. The blue points represent the points of intersection of the plane in one direction, and the red points the other. The one-D map is in the top right hand corner. The set of points (red or blue) that are plotted in the one-D map may be chosen by pressing or depressing the 'Plot Red Points' clickbox. The slope of the one-D map can be calculated by pressing the 'Calculate Slope' button.

Note: ABC++ works best with MATLAB R13 version 6.5

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